IMEN891K / IDSC501 / AXGS-501 · DATA ENGINEERING · FALL 2026

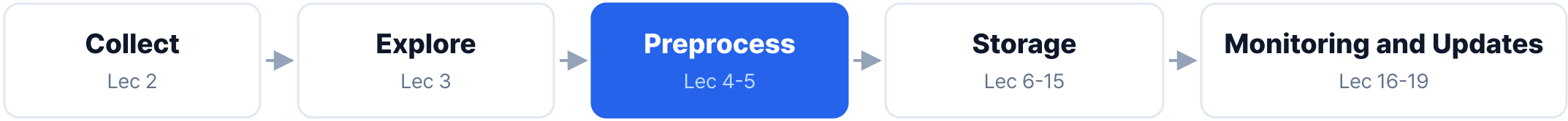
4.

Pre-processing I: Cleaning & Integration

Sangwoo Mo · POSTECH
Industrial & Management Engineering

Where We Are

PART I · Collection and Preparation | **PART II** | **PART III**

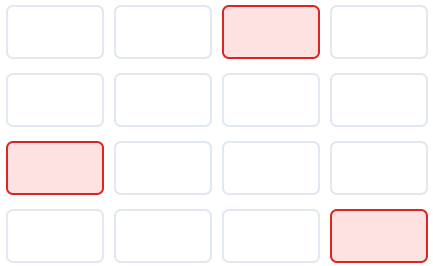


nothing downstream can recover what preprocessing throws away

Outline: Four Stages of Preprocessing

01

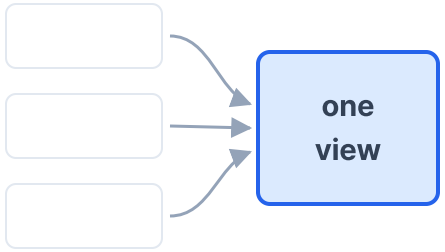
Cleaning



Wrong values in one table

02

Integration



Tables that disagree

03

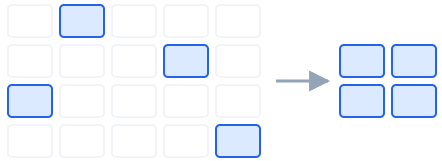
Transformation



Values a model can read

04

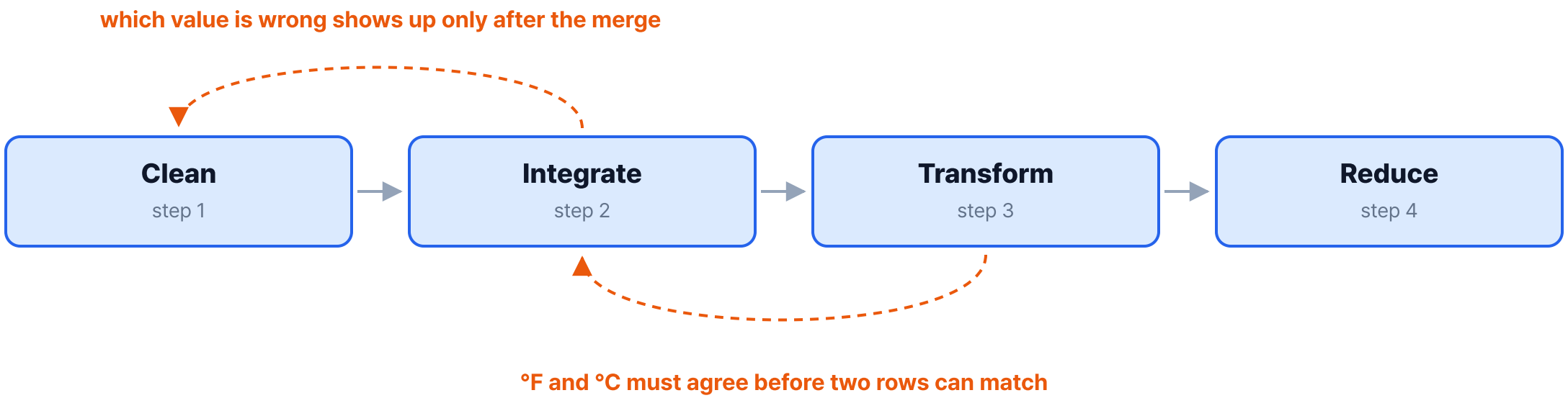
Reduction



Less data, same signal

Not a One-Way Pipeline

Two of today's sections are steps running out of order

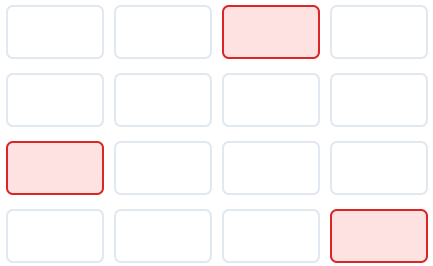


Both orange arrows run backward, and both are sections of part 2

Cleaning

01

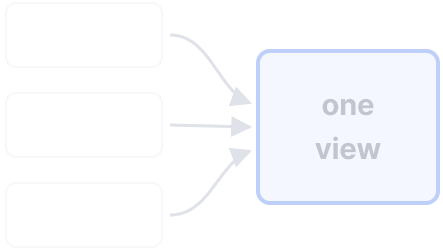
Cleaning



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Tables that disagree

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Transformation



Values a model can read

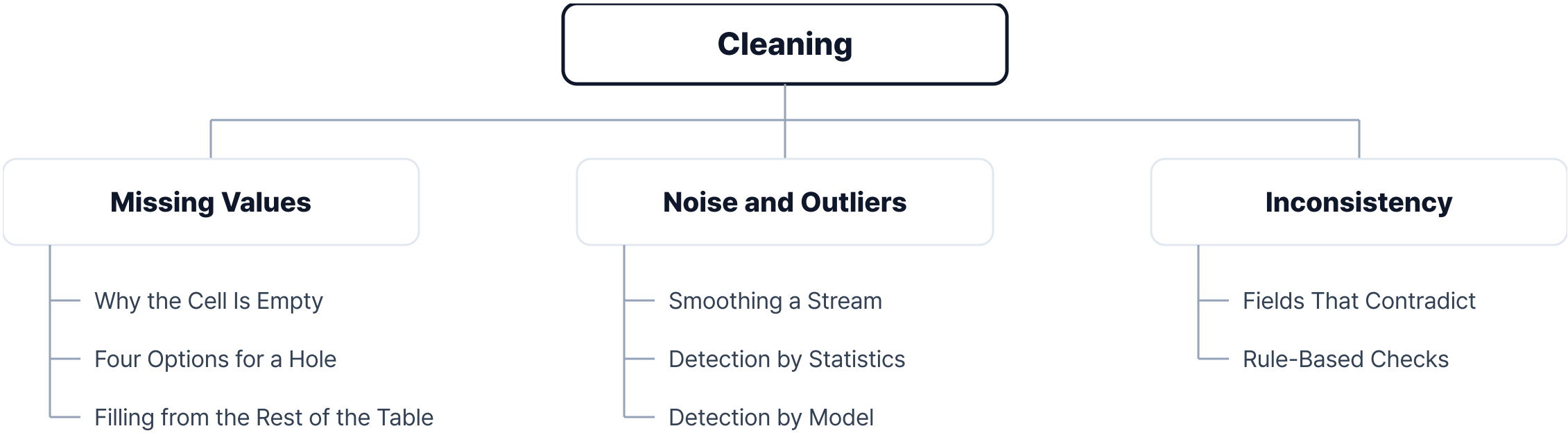
04

Reduction

Less data, same signal

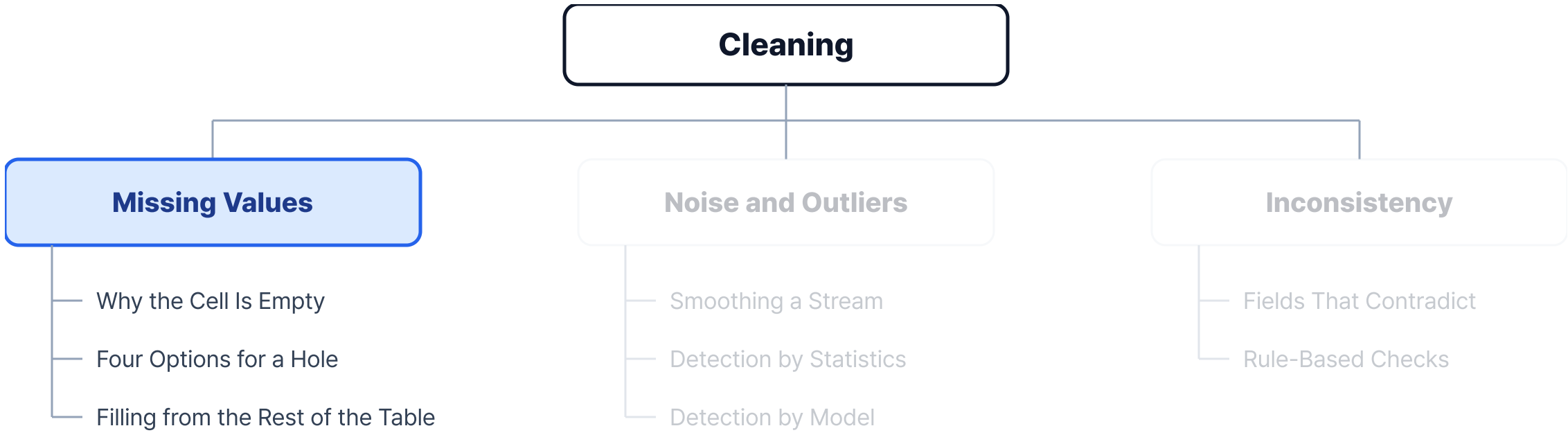
Three Branches of Cleaning

Missing, wrong, and contradicting values are three different problems



Calling all three ‘dirty data’ is how good rows get deleted

Missing Values



Why the Cell Is Empty

ROW	STATE	SALARY	EXPERIENCE
1	NY	57,400	Mid
2	TX	**missing**	Entry
3	NJ	90,000	High
4	VT	36,900	Entry
5	TX	**missing**	Mid
6	CA	76,600	High
7	NY	85,000	High
8	CA	**missing**	Entry

Three empty cells can have **three different reasons**, and the remedy depends on which one it was

Caveat: Why is it missing?

Holes independent of the value

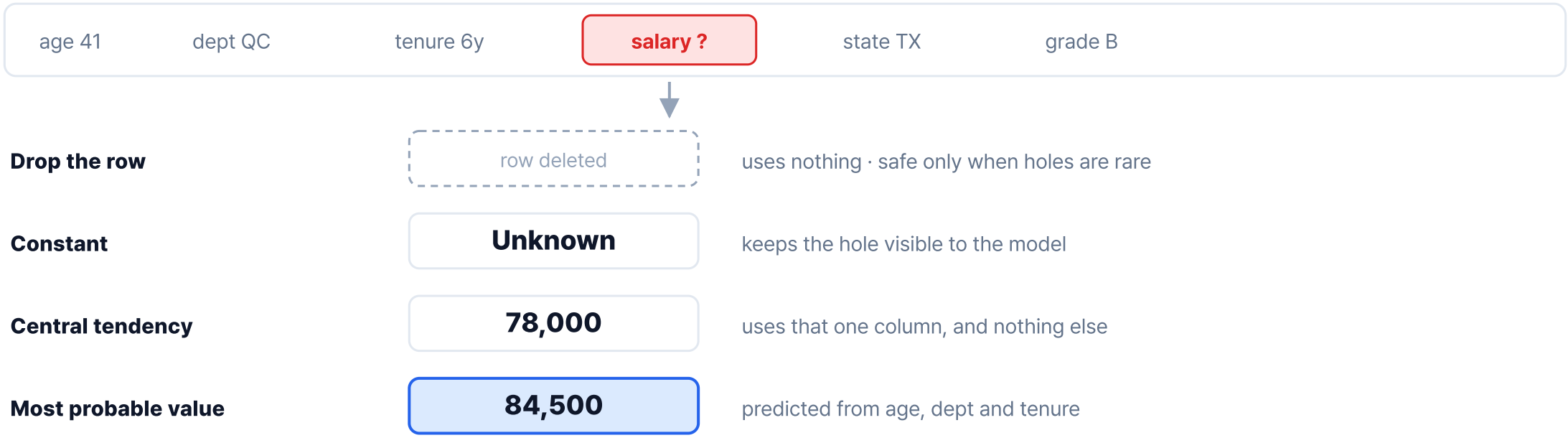
age	25	26	29	30	30	31	44	46	48	51	52	54
IQ	×	121	91	×	110	×	118	93	×	×	116	×

Holes concentrated in the younger half

age	25	26	29	30	30	31	44	46	48	51	52	54
IQ	×	×	×	×	×	×	118	93	116	141	97	104

Below, no young IQ was ever observed. **No imputer fixes this.** The question is not which method to use but why those rows are missing, and that answer lives in how the data was collected

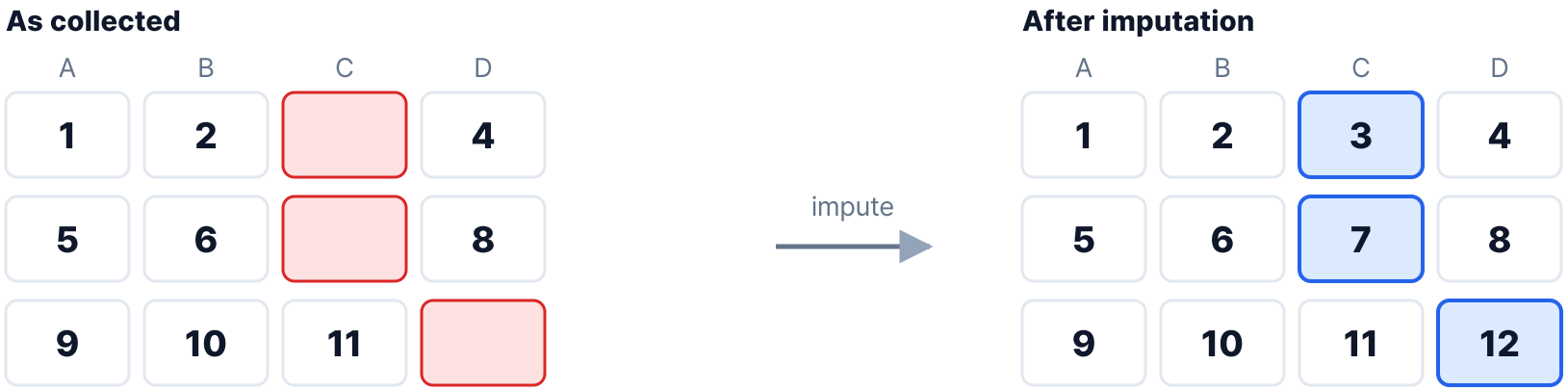
Four Options for a Hole



Top to bottom, **each one uses more of the rest of the record**

Imputation as Estimation

Filling a hole from the rest of the data

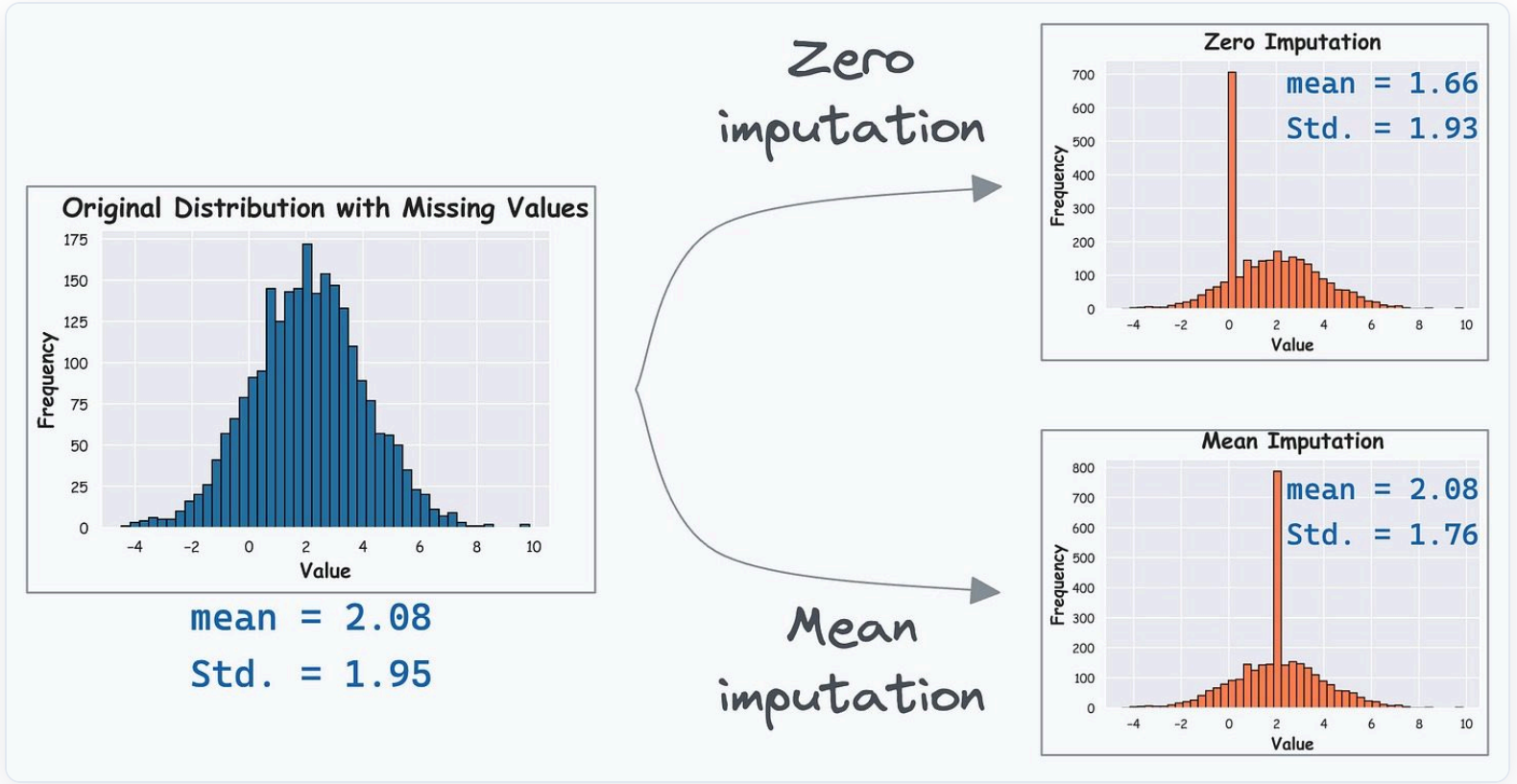


$\hat{x}_{ij} = f(\ X_{observed}\ )$ every filled cell is a prediction from the rest of the table

Every filled cell is an estimate, not an observation

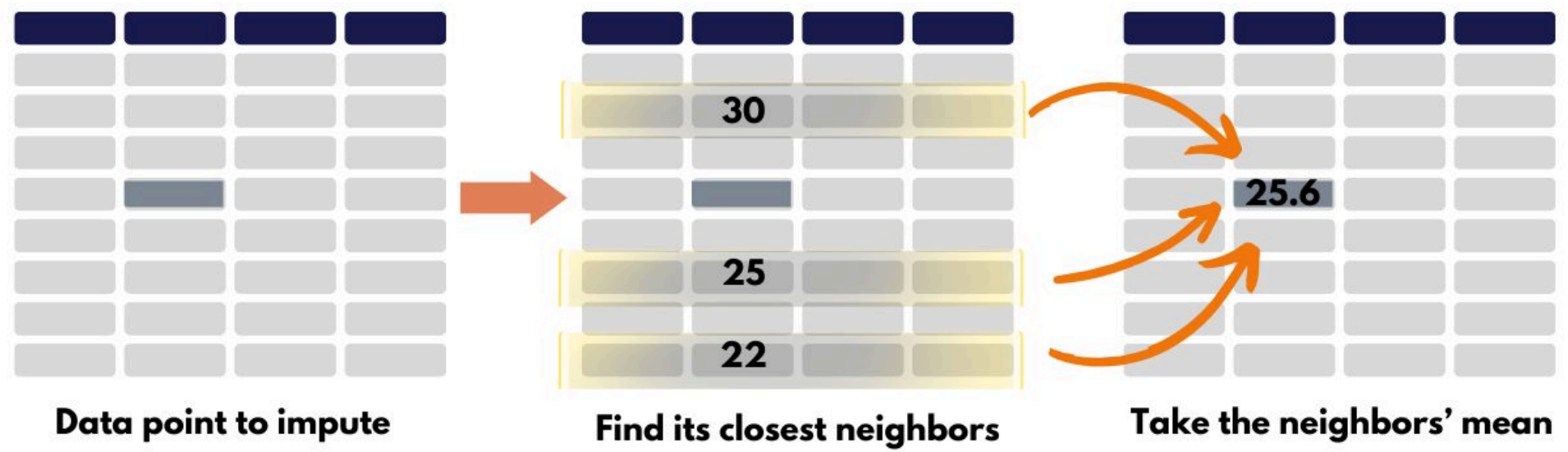
Caveat: Mean Imputation

Every hole gets the same value, and the spread of the column collapses



Mean imputation holds the mean at 2.08 but **drops the standard deviation to 1.76 and leaves a spike**

k-Nearest Neighbor (kNN) Imputation



$$\hat{x}_{ij} = (1/k) \sum_{r \in N_k(i)} x_{rj}$$ closest in the other columns, or in an embedding of them

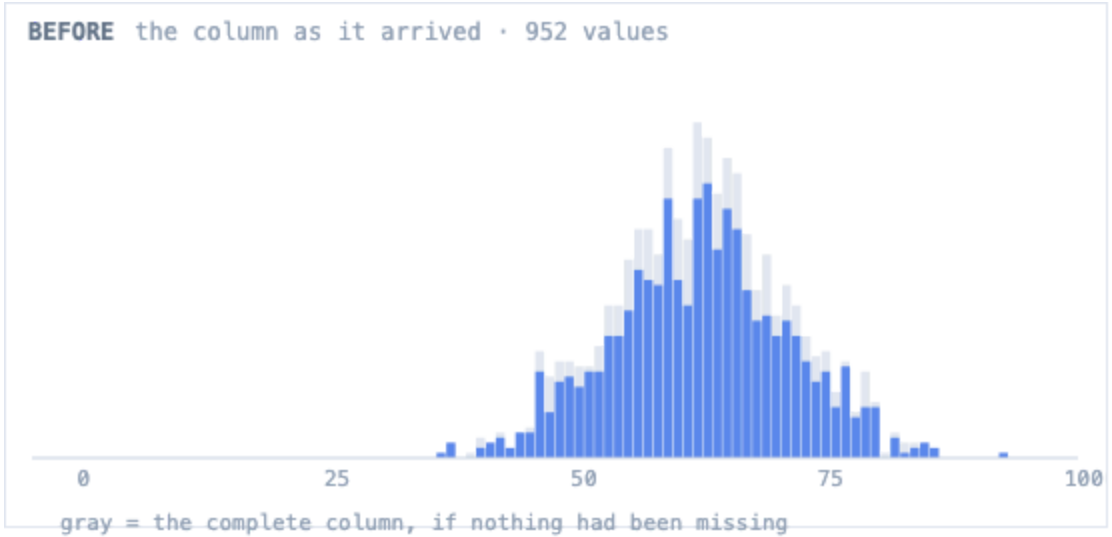
Average the k most similar rows. **Push k to every observed row and you are back at mean imputation**

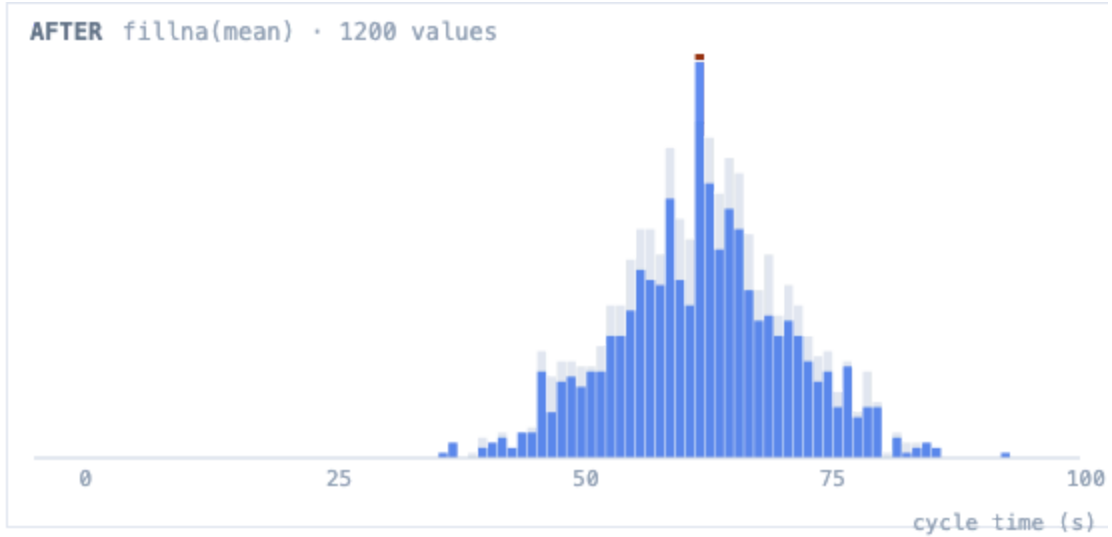
Demo: Imputation Methods

One column of cycle times, 1,200 rows: every fill changes the column

1. Compare mean and kNN fill. 2. What changes when the helper column contains little information?

Missing 20% Helper column ρ 0.75 Method fillna(mean)





true **μ 61.7 σ 8.9** · after **μ 61.7 σ 7.9** · rows **1200**

KNNImputer fills a hole from rows that are close in a second column. ρ is how much that column knows about this one. At ρ 0 it knows nothing.

spread −11% · mean −0.0

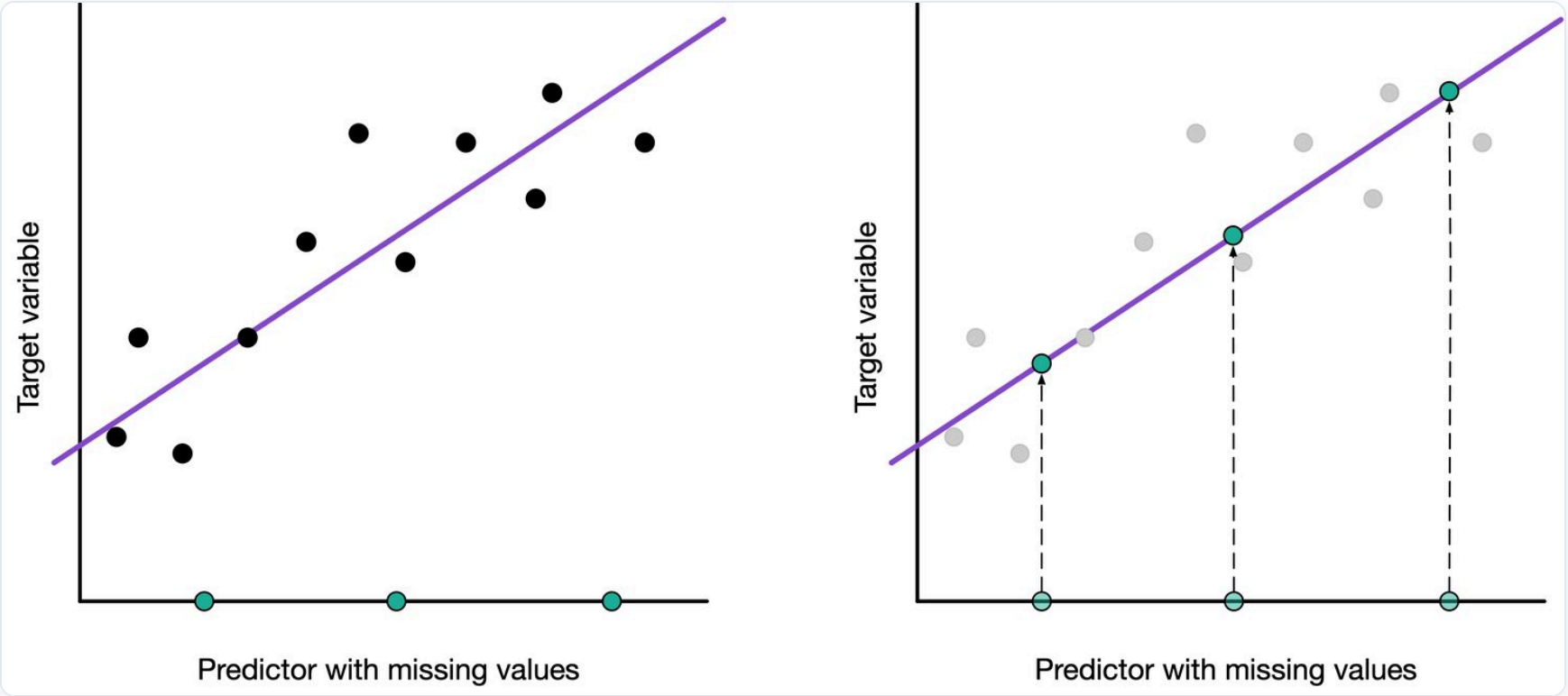
Answer: Imputation Choices

Compare distributions, not only the filled cells

Choice	What to inspect
`Mean fill`	`Mean can stay close; variance shrinks`
`kNN fill`	`Does the helper column contain information?`
`Missingness mask`	`Keep observed and estimated cells distinguishable`

A filled cell is an estimate, not a new observation

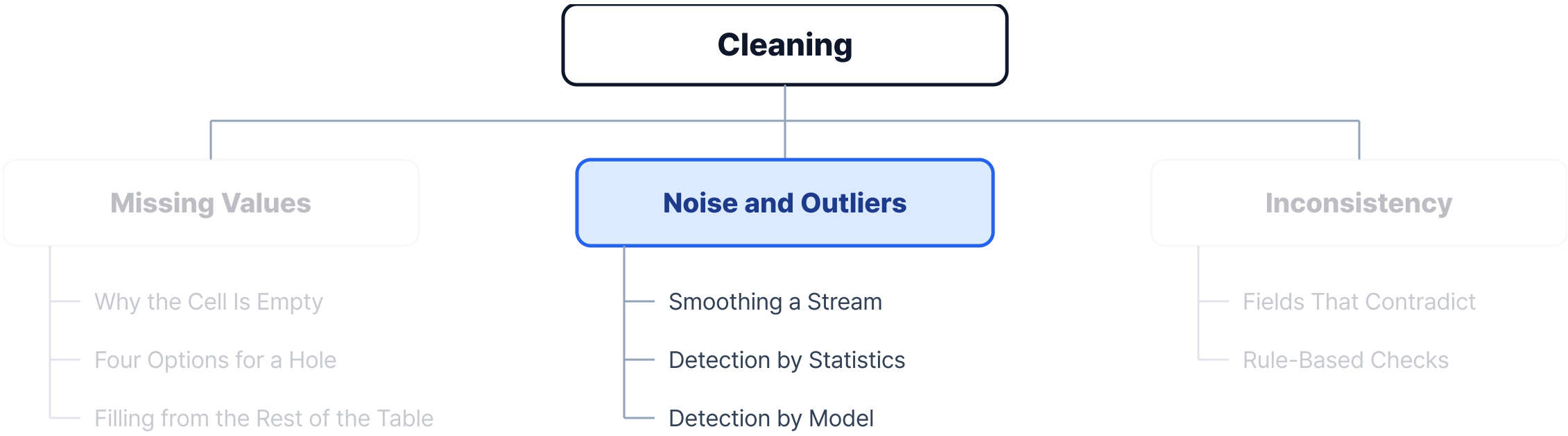
Regression Imputation



$$\hat{x}_{ij} = b_0 + \sum_{m \neq j} b_m x_{im}$$ no residual is added back, so the spread shrinks

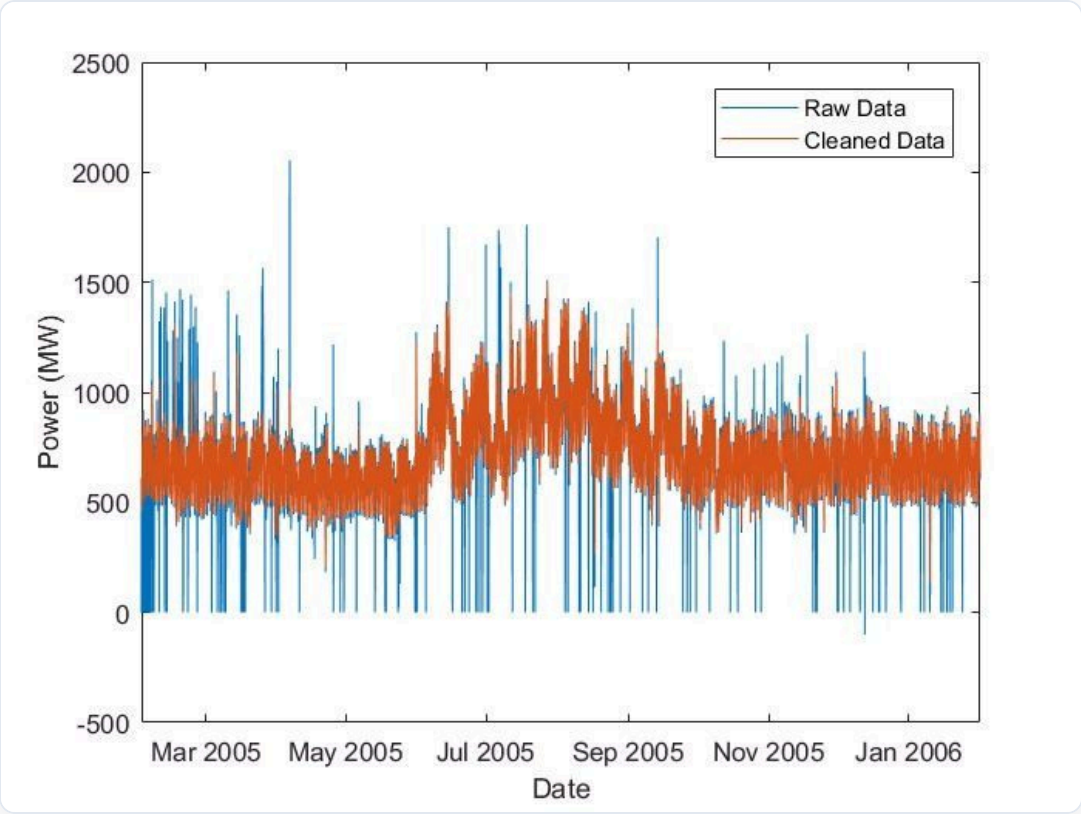
The green points sit exactly on the line. **The residual is gone, so the correlation looks stronger than it is**

Noise and Outliers



Raw Sensor Streams

Metered power demand, with two different kinds of error mixed in

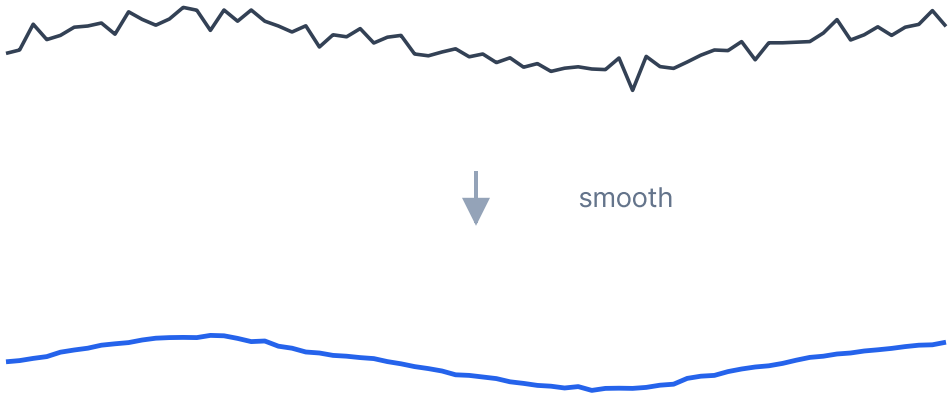


Blue spikes drop to zero and below. **That is the meter dropping out, not demand**

Noise vs Outliers

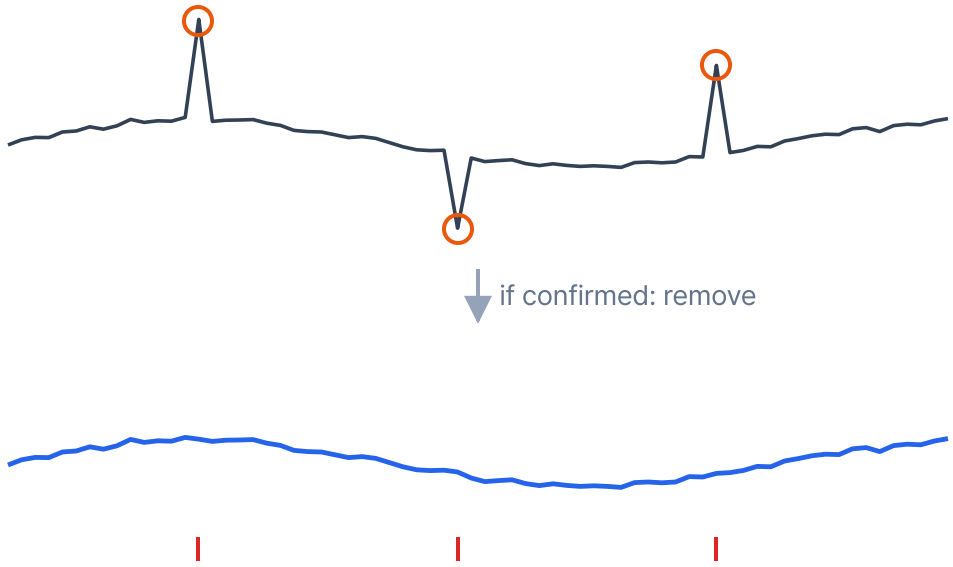
Smooth measurement noise; investigate outliers before changing them

Noise: every value is a little wrong



all 1,200 rows survive

Outliers: a few values sit far away

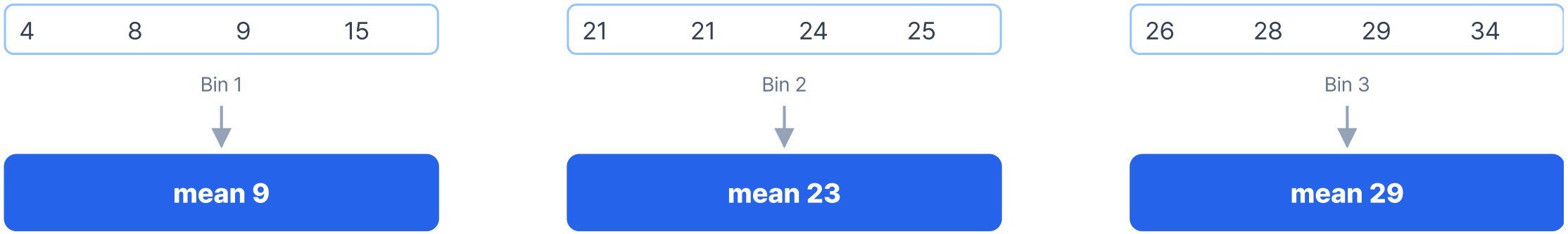


three rows are gone

Smoothing changes **every value and keeps every row**. Removal keeps the values and **drops rows**

Smoothing by Binning

Sorted values (price, $)



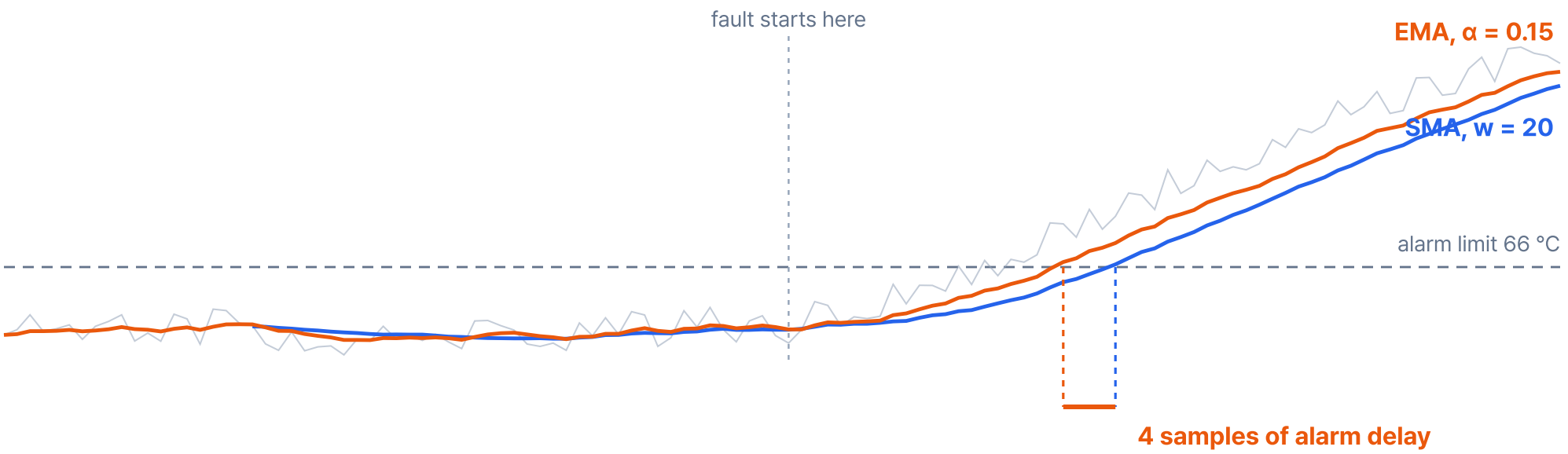
Smoothed

`9, 9, 9, 9, 23, 23, 23, 23, 29, 29, 29, 29`

Twelve values become three distinct ones. **Smoothing buys steadiness with resolution**

Moving Averages for Streams

Simple (SMA) and exponential (EMA) moving averages use only current and past samples



$\hat{x}_t = (1/w) \sum_{k=0}^{w-1} x_{t-k}$ · $s_t = \alpha\, x_t + (1 - \alpha)\, s_{t-1}$ SMA weights the window equally; EMA uses one α

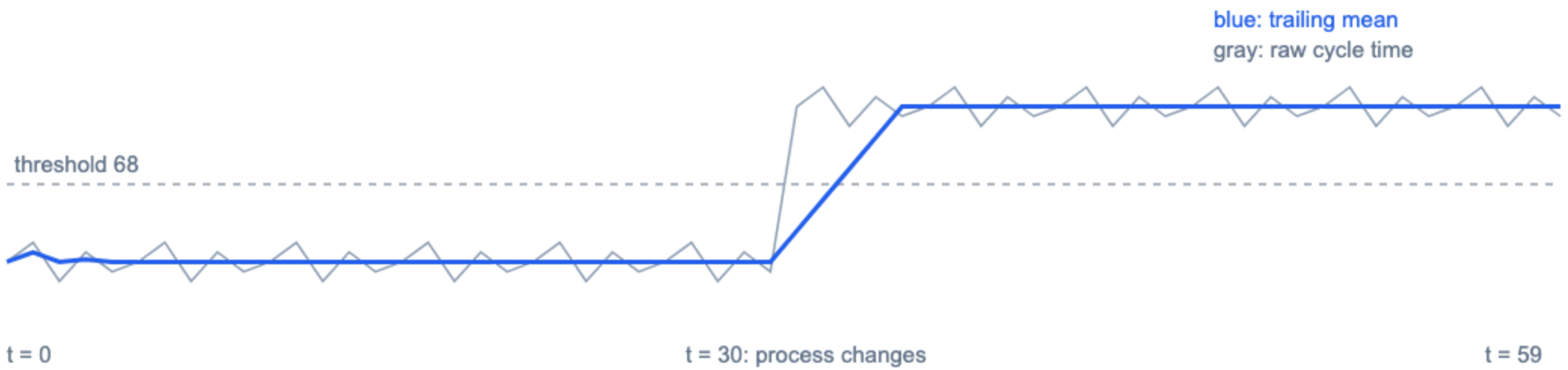
Both windows look only backward. **EMA crosses four samples before SMA, and that gap is your alarm delay**

Demo: Smoothing a Stream

Current and past cycle times (seconds); wider trailing windows respond later

1. Compare windows 1, 5 and 11. 2. When does the smoothed signal cross 68 s?

Window 5



`window 5 · first mean ≥ 68 at t = 32 · delay 2 samples`

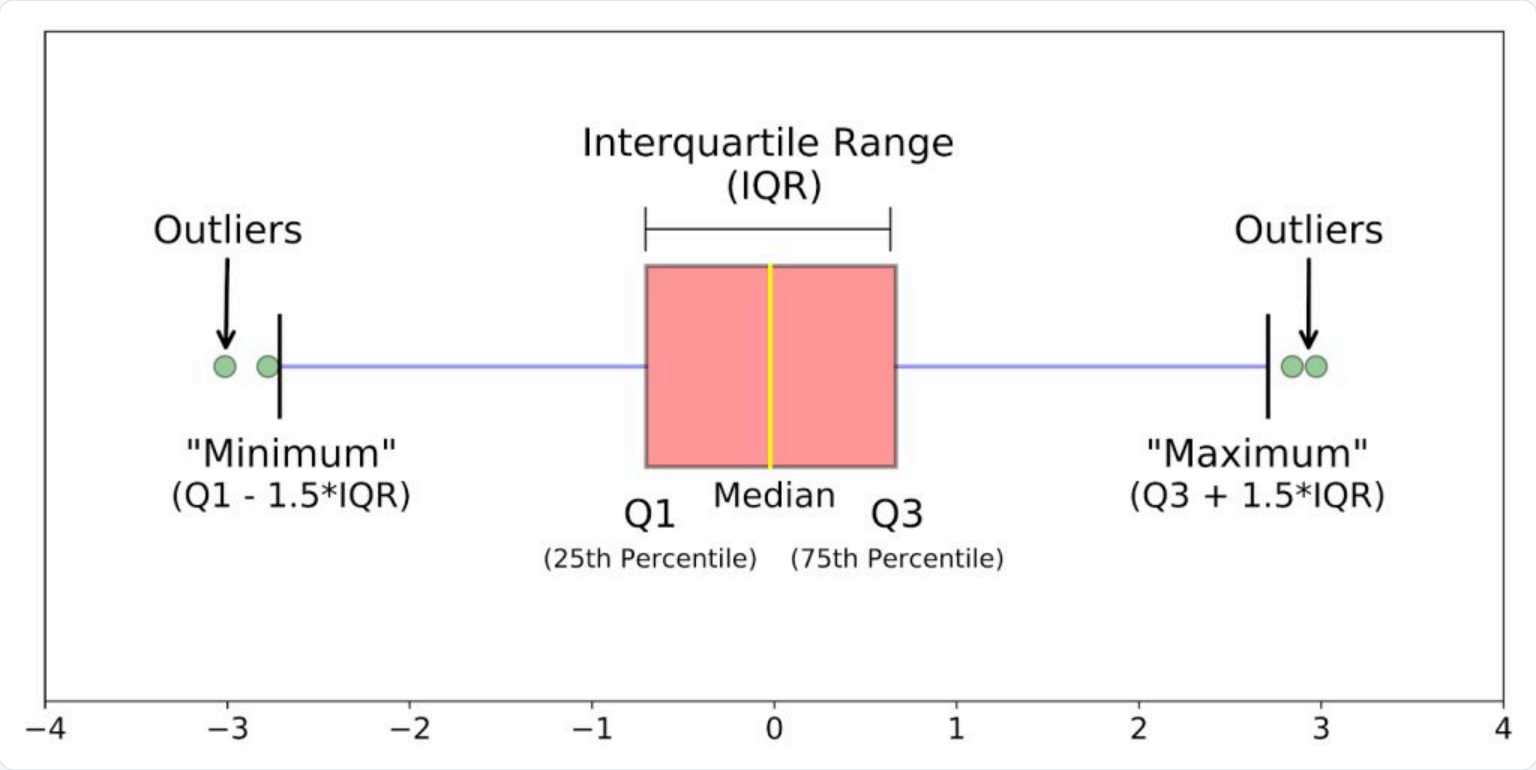
Answer: Smoothing Delay

A larger trailing window needs more new observations

Window	First smoothed value ≥ 68 s	Delay after t = 30
1	30	0 samples
5	32	2 samples
11	35	5 samples

Less fluctuation comes with a later response

Outlier Detection by Statistics



$z_i = (x_i - \bar{x}) / s$ or outside $Q_1 - 1.5 \cdot IQR$, $Q_3 + 1.5 \cdot IQR$ the z-score uses the mean; the fence uses only quartiles

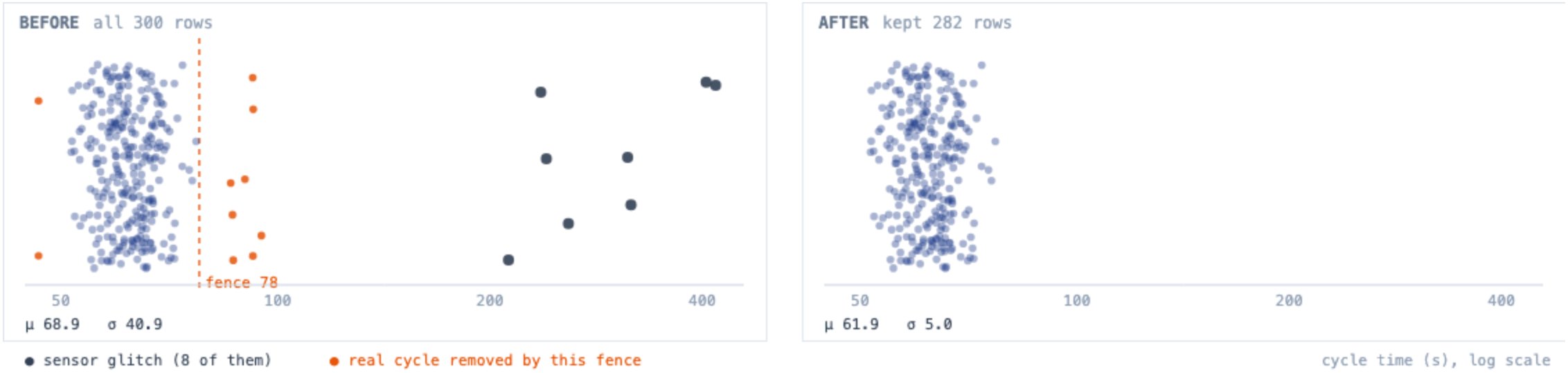
Past 1.5 × IQR from the box. **The rule is univariate: 40 kg and 190 cm each pass their own fence**

Demo: IQR Fence Width

300 cycle times with eight sensor glitches and eight genuinely slow cycles

1. Change the fence width. 2. Which genuine cycles disappear along with sensor glitches?

Fence width k 1.5 drop anything outside Q1 − k·IQR … Q3 + k·IQR



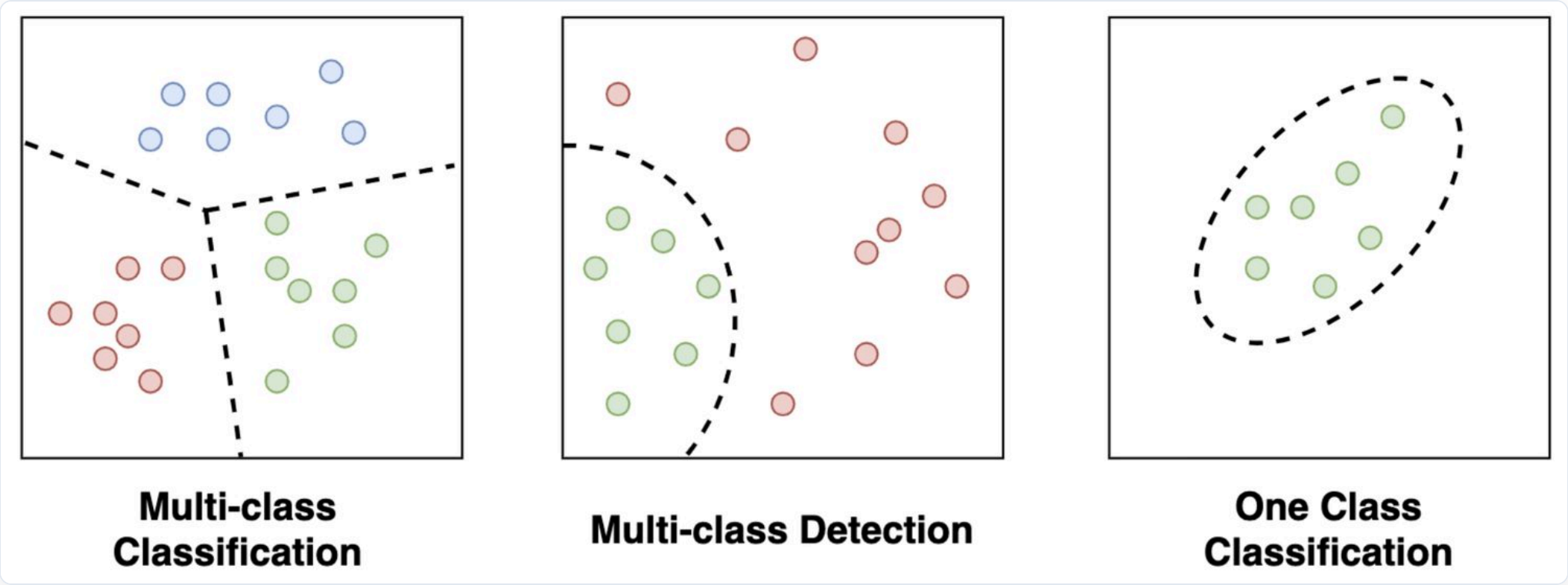
fence at Q3 + 1.5 × IQR = **78 s** · removed **18** rows · σ **40.9 → 5.0**

1.5 is Tukey's convention, not a test. Pick k by what the fence deletes, and report the k you used.

10 real cycles deleted with them

One-Class Classification

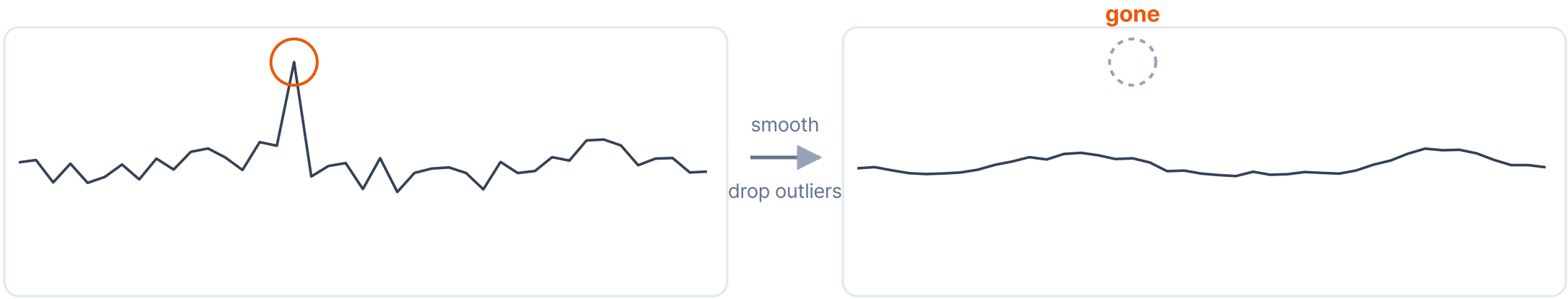
Draw a boundary around normal and flag what falls outside



A fence per column misses unlikely combinations. **Learn one boundary around normal rows instead**, from normal data alone

Caveat: Irreversible Cleaning

Cleaning cannot be undone downstream



As recorded

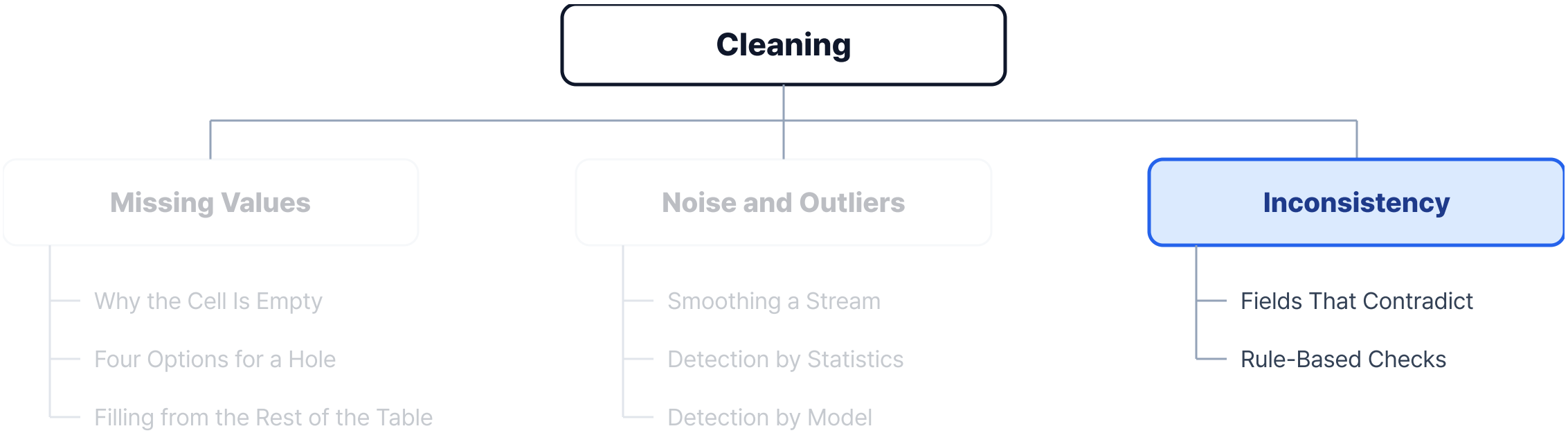
one spike at 14:02

As stored

nothing marks where it was

The spike is not hidden in the stored column. **It is gone, and so is the question**

Inconsistency



Fields That Contradict Each Other

ORDER	ORDERED	SHIPPED	QTY	UNIT	AMOUNT
A-1042	2026-03-05	**2026-03-01**	12	kg	480,000
A-1043	2026-03-06	2026-03-09	**0**	kg	**1,250,000**
A-1044	2026-03-06	2026-03-08	26	**lb**	1,040,000
A-1045	2026-03-07	2026-03-10	15	kg	600,000

Nothing is missing here. Each value is wrong only next to another value

Rule-Based Checks

Domain knowledge determines which consistency rules are valid

CHECK	ROW IT STOPS	WHERE THE RULE COMES FROM
shipped ≥ ordered	A-1042	How the business works
qty = 0 → amount = 0	A-1043	How the business works
unit ∈ { kg }	A-1044	A decision someone has to make

Three checks, three rows stopped. A model trained on this table would have learned the errors as normal

Demo: Checking Consistency

Apply two domain rules to the recorded fields

run_id	started	ended	total	defects
R-01	09:00	09:20	100	4
R-02	09:30	09:10	80	2
R-03	10:00	10:15	20	24

1. Which rows violate time or count rules?
2. Can the rule tell you which cell is wrong?

Answer: Violations and Repairs

A failed check identifies a contradiction, not the correct replacement

run_id	Violated rule	Action
R-01	none	keep
R-02	ended ≥ started	check source timestamps
R-03	0 ≤ defects ≤ total	check count definitions

Flag contradictions before overwriting recorded values

Integration

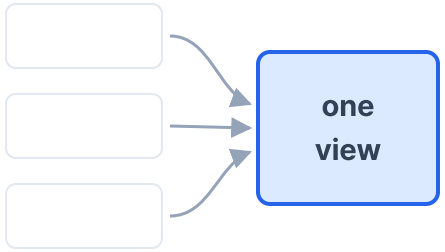
01

Cleaning

Wrong values in one table

02

Integration



Tables that disagree

03

Transformation



Values a model can read

04

Reduction

Less data, same signal

One Customer, Three Systems

Customer relationship management (CRM), supply chain management (SCM), and enterprise resource planning (ERP)

SYSTEM	OWNED BY	KEY	NAME AS STORED	REVENUE FIELD
CRM	Sales	C-10482	Hanil Precision Co., Ltd.	booked_amount
SCM	Logistics	V0000734	HANIL PRECISION	(none)
ERP	Finance	2010-HP	한일정밀(주)	net_sales

No shared key, no shared spelling. Nothing joins these three rows on its own

Three Questions of Integration

Columns first, then rows, then values

Integration

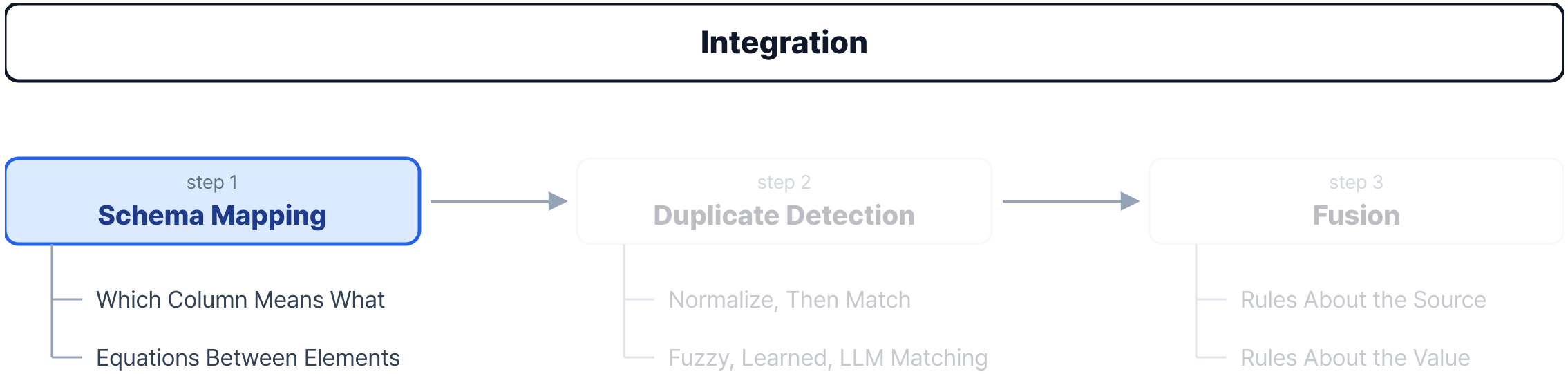
step 1
Schema Mapping
- Which Column Means What
- Equations Between Elements

step 2
Duplicate Detection
- Normalize, Then Match
- Fuzzy, Learned, LLM Matching

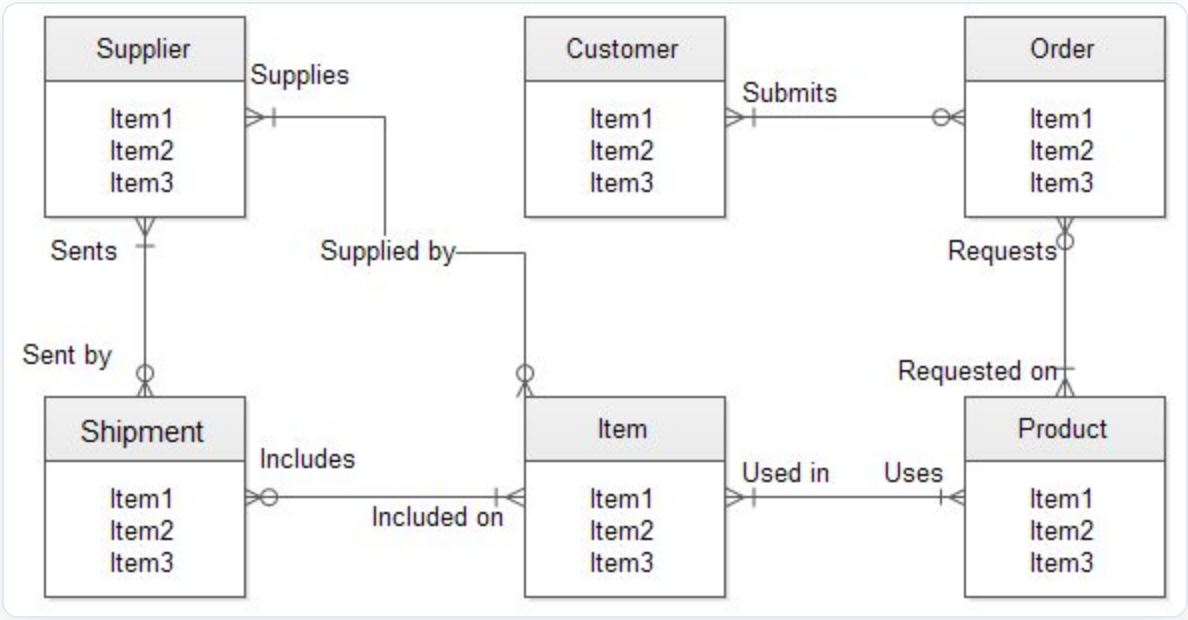
step 3
Fusion
- Rules About the Source
- Rules About the Value

Skip a step and the join silently answers the wrong question

Schema Mapping

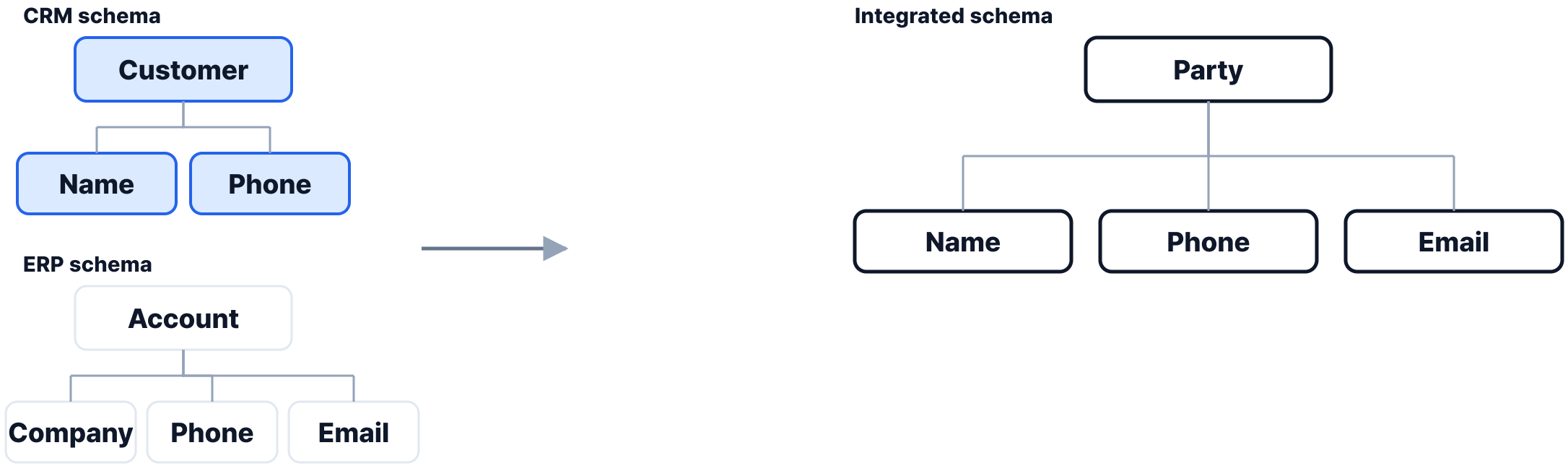


Same Name, Different Entity



Here **Order and Shipment are separate entities**. Another system may keep both in one table

Mapping Two Schemas into One



Matched elements collapse into one. **What only one side has still has to fit somewhere**

Demo: Mapping Customer Records

Match meaning before matching rows

Source field	Target field	Transform or decision
`CRM.Name / ERP.Company`	`Party.name`	`retain name variants`
`CRM.customer_id / ERP.account_id`	`source_id`	`keep source + original ID`
`CRM.booked_amount / ERP.net_sales`	`separate`	`different measures`

1. Do booked_amount and net_sales belong in one column?
2. Do matching column names prove matching customers?

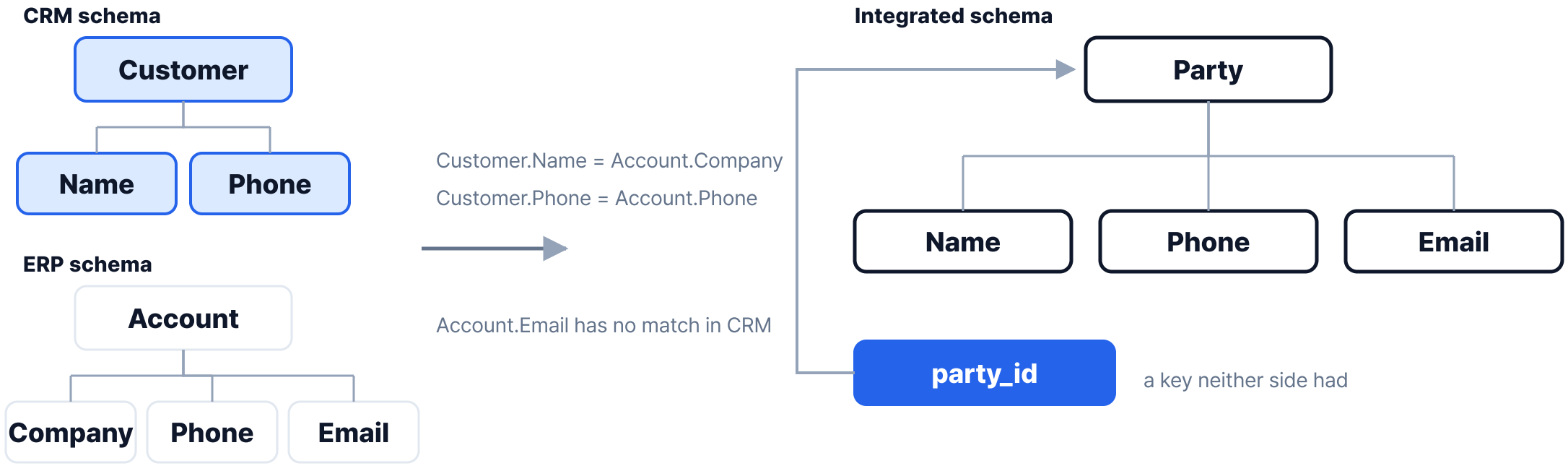
Answer: A Shared Schema

Align field meanings while preserving different measures and source identifiers

Target field	Source mapping	Meaning
`name + aliases`	`CRM.Name / ERP.Company`	`organization names`
`source / source_id`	`CRM ID / SCM ID / ERP ID`	`original identifiers`
`booked_amount`	`CRM.booked_amount`	`booked orders`
`net_sales`	`ERP.net_sales`	`recognized net sales`

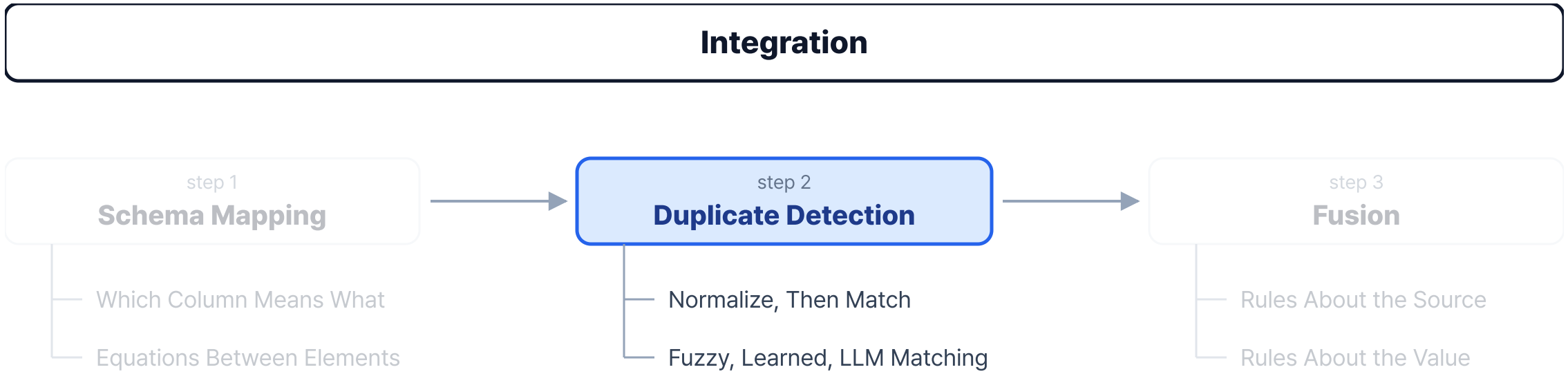
Mapping columns does not establish that two rows refer to the same entity

Equations Between Elements



Each equation merges two elements. **A new key appears when neither side's key works**

Duplicate Detection



Demo: Exact Duplicates

Normalize, then drop_duplicates: twelve supplier rows, six real suppliers

1. Normalize until six suppliers remain. 2. What breaks when you keep only the first word?

☐ lowercase ☐ strip punctuation ☐ collapse spaces ☐ drop Co / Ltd / Inc ☐ keep the first word `12 rows → 12 · still 6 too many`

BEFORE: 12 ROWS AS STORED	AFTER: THE KEY IT NORMALIZES TO	KEPT
Samsung Electronics Co., Ltd.	`Samsung Electronics Co., Ltd.`	keep
samsung electronics co ltd	`samsung electronics co ltd`	keep
Samsung Electronics Co., Ltd.	`Samsung  Electronics Co., Ltd.`	keep
Samsung Electronics	`Samsung Electronics`	keep
LG Electronics Inc.	`LG Electronics Inc.`	keep
LG ELECTRONICS INC.	`LG ELECTRONICS INC.`	keep
LG Electronics	`LG Electronics`	keep
Hyundai Mobis Co., Ltd.	`Hyundai Mobis Co., Ltd.`	keep
Hyundai Mobis	`Hyundai Mobis`	keep
POSCO Holdings	`POSCO Holdings`	keep
POSCO International Corp.	`POSCO International Corp.`	keep
Samsung Heavy Industries Co., Ltd.	`Samsung Heavy Industries Co., Ltd.`	keep

Normalize before you dedup, never after. Casing, whitespace and punctuation account for most byte-level mismatches.

Answer: Normalization and Merging

A shorter string can accidentally erase a real distinction

Normalization	Result in this example
Case, punctuation, spaces, suffixes	12 rows → 6 suppliers
Keep only the first word	Distinct Samsung and POSCO entities merge
Preserve original name and source ID	The decision remains auditable

Normalization should support matching without hiding the original record

Fuzzy Matching

Each field gets a similarity score, then one weighted total against a cutoff



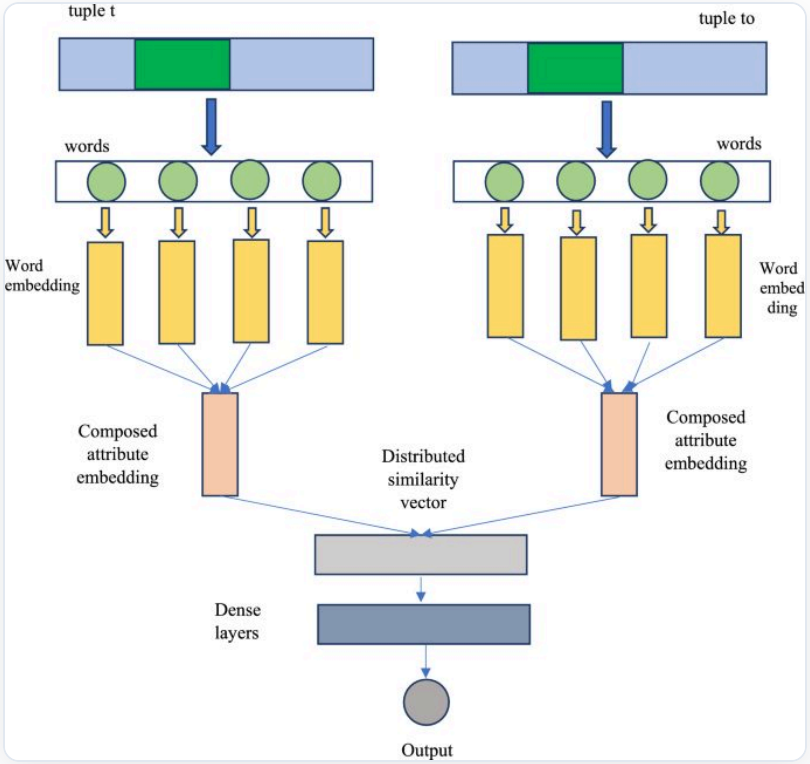
$$\text{score} = \sum_f w_f \cdot \text{sim}_f(a, b) \qquad \text{sim} = 1 - \text{edit}(a,b) \,/\, \max(|a|,|b|)$$

on lowercased text, then one weighted total

Per-field scores become one number. **The cutoff you pick trades false merges against missed ones**

Learned Matching

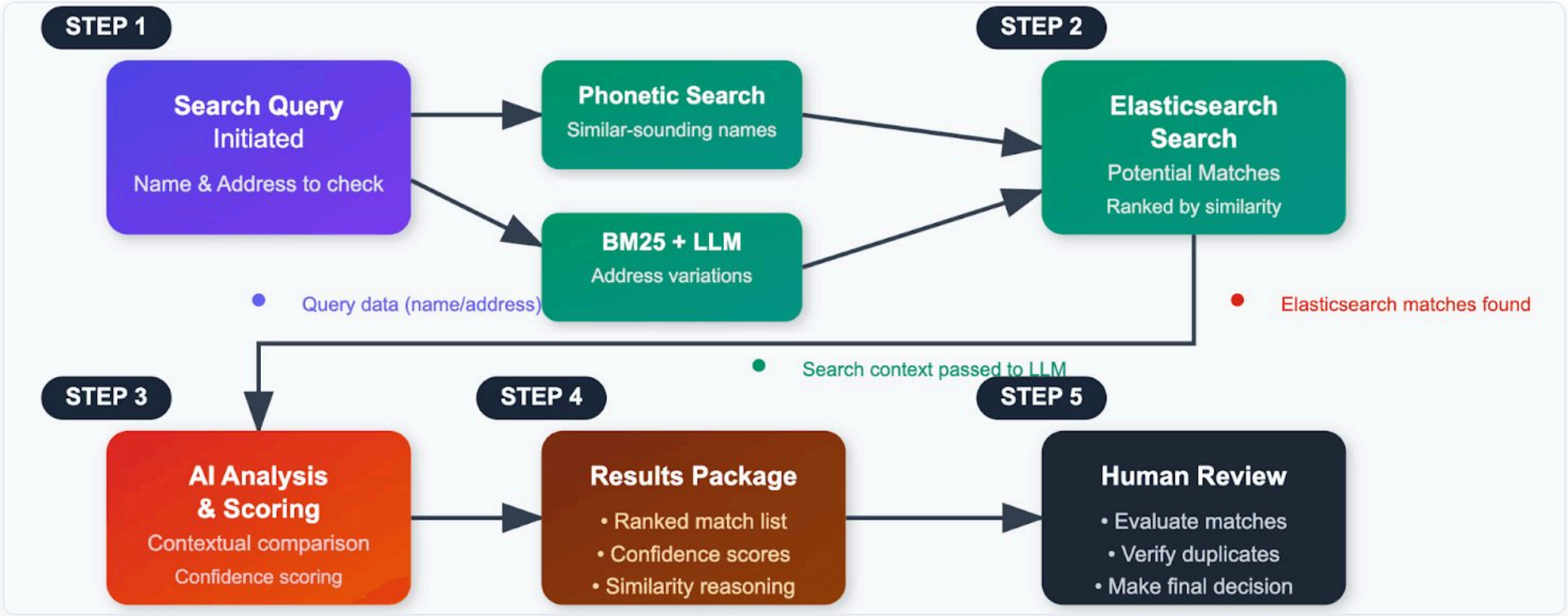
A trained matcher sees past the spelling



'삼성전자' and 'Samsung Electronics' share no character, so edit similarity scores them zero. **A multilingual embedding puts them close**

LLM-Assisted Matching

Large language models (LLMs) propose the matches, humans verify them



Cheap search first, model second, person last. **Drop the last box and wrong merges pile up unseen**

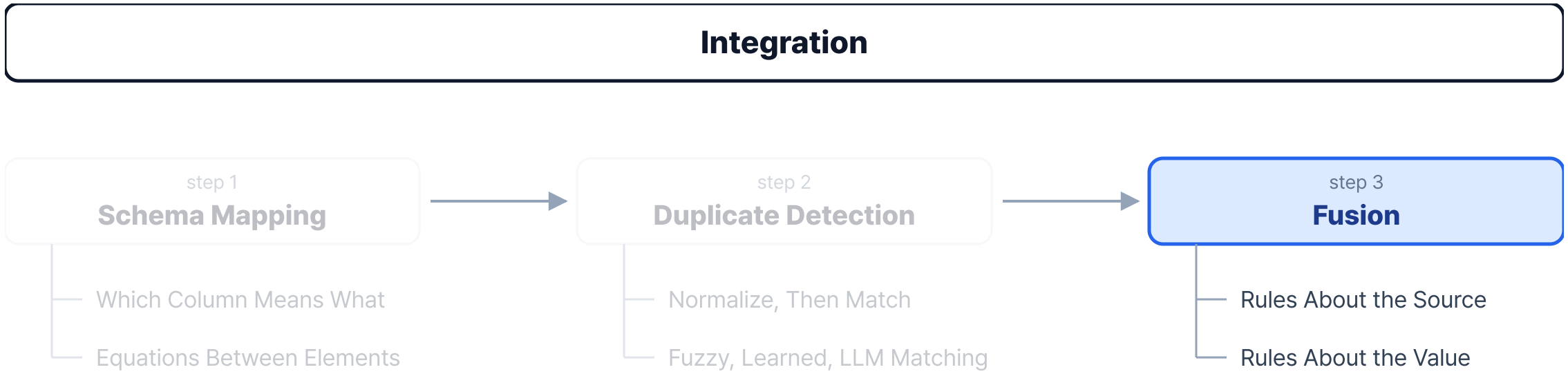
Answer: Source IDs and Party IDs

A verified link groups records without rewriting their original identifiers

source	source_id	party_id	basis
CRM	C-10482	P-073	verified registration ID
SCM	V0000734	P-073	verified registration ID
ERP	2010-HP	P-073	verified registration ID

Similar names generate candidates; verified evidence supports a merge

Fusion


Integration
step 1
Schema Mapping
Which Column Means What
Equations Between Elements
step 2
Duplicate Detection
Normalize, Then Match
Fuzzy, Learned, LLM Matching
step 3
Fusion
Rules About the Source
Rules About the Value

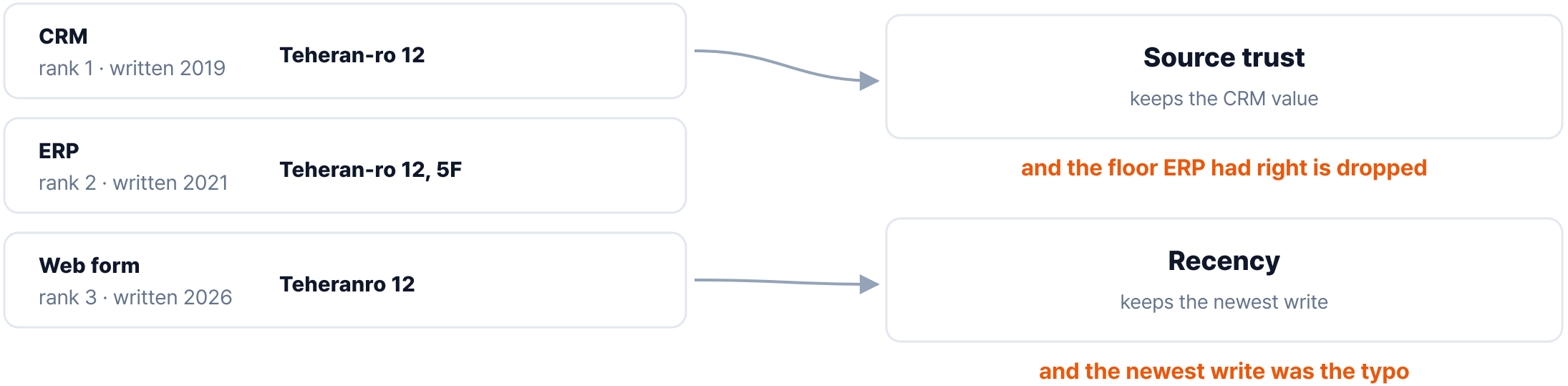
Conflicts After Matching

Matched rows still disagree on values

	name	address	phone
CRM	**Hanil Precision**	**Teheran-ro 12**	**02-555-0100**
SCM	**Hanil Precision**	**Teheran-ro 12**	**02-555-0100**
ERP	**Hanil Precision**	**Teheran-ro 12, 5F**	**02-555-0199**
Web form	**Hanil Precision**	**Teheranro 12**	**010-2233-4455**

Same customer, three spellings of one address, three phone numbers

Rules About the Source



Neither rule reads the value. **One keeps a stale address, the other keeps a typo**

Rules About the Value

Majority voting

CRM	**02-555-0100**
copy	**02-555-0100**
report	**02-555-0100**
ERP	**02-555-0199**

three agree, but two of them are copies of the first

Learned

conflicts a person already settled

Seoul vs SEOUL → a person kept Seoul

02-555-0100 vs 0199 → a person kept 0100

Model

picks the winner of the next conflict

$$\hat{x} = \text{argmax}_v \sum_k w_k \cdot 1[x_k = v]$$ weighted vote: w_k is the trust in source k

Three copies of one value **count as one vote, not three**

Demo: Resolving a Customer Conflict

Choose a canonical value and keep the evidence

Field	Candidate values	Evidence
name	Hanil Precision / 한일정밀(주)	same verified registration ID
address	12 / 12, 5F / Teheranro 12	verified delivery: 12, 5F
phone	02-555-0100 / 02-555-0199 / 010-2233-4455	no fresh confirmation

1. Which values can be settled now?
2. Which field needs a review instead of a vote?

Answer: A Canonical Customer

Publish resolved values while keeping unresolved conflicts visible

party_id	field	canonical value	evidence / status
P-073	name	Hanil Precision	verified ID; aliases retained
P-073	address	Teheran-ro 12, 5F	verified delivery + ERP
P-073	phone	NULL	review; candidates retained

A canonical record includes provenance and unresolved decisions

Takeaway: Cleaning decides what is true

Three things to carry out of today



A cleaned value is a guess

Imputation and smoothing replace what was recorded. Keep a mask column and the raw copy



Some errors only show up between fields

Statistics rarely flag them. You write the check from how the business works



Merging is three decisions, not a join

Columns, then rows, then which value wins